

Cover Letter — PSB 2027 Submission

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Beyond Top- k : Dimension-Resolved Explainability in Protein Language Model Embeddings

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Protein language models such as ESM2 encode rich structural signal, yet embedding-based similarity search typically returns a ranked neighbor list without explaining *why* two proteins are close in vector space. We present **XQdrant**, an explainable retrieval layer that decomposes cosine similarity into the specific embedding dimensions driving a match (`dims_explained`). On a SCOP-stratified corpus of 22,118 chains (1,500 held-out queries), XQdrant achieves fold-level recall@1 of 0.898 and mAP of 0.896; `dims_explained` attributions are post-hoc and leave Qdrant neighbor rankings unchanged. Permutation tests on 10,235 retrieved pairs show fold-matched neighbors score higher than mismatched pairs (Mann–Whitney $p < 10^{-57}$; label-permutation gap $p = 0.0002$; AUROC = 0.63). Active-site localization on 1,590 SwissProt-annotated pairs shows per-residue attribution enrichment at UniProt sites ($p \approx 0$). A thermostability case study ($n = 44$ ortholog pairs) links attributions to ion-pair density ($\rho = -0.30$, $p = 0.052$). Workflow timing: indexed attributed search median 0.074 s vs. Foldseek+TM-align 4.2 s automated (+ estimated 10 min PyMOL), with zero manual steps for rationale.

Keywords: protein language models; explainable AI; vector retrieval; structural bioinformatics; ESM2.

1. Introduction

Embedding-based protein search has become a practical alternative to sequence and structure alignments for large-scale corpus screening. Methods built on protein language models (PLMs) such as ESM2¹ represent chains as fixed-length vectors amenable to fast approximate nearest-neighbor indexes. However, standard retrieval returns only a similarity score and neighbor identity. For mechanistic biology—enzyme engineering, thermostability engineering, or functional annotation—researchers need to know whether a match reflects shared catalytic architecture, hydrophobic core packing, or incidental sequence similarity.

Explainability in PLM embeddings remains limited. We address this gap with **XQdrant**, a retrieval system that exposes `dims_explained`: the top- N embedding dimensions and non-negative contributions underlying each cosine-similarity match in Qdrant.

Our contributions are:

- (1) An end-to-end pipeline from PDB structures to ESM2 embeddings, Qdrant indexing, and dimension-resolved attribution (Fig. 1).
- (2) A train-only dim→profile map linking attributed dimensions to DSSP binned structural

profiles, with row- and label-permutation null models.

- (3) Empirical validation across retrieval (A), attribution specificity (B–C), thermostability proxies (D), active-site localization (E), and analyst workflow timing (F).
- (4) Ablations over ESM2 layer depth, pooling strategy, and top- N attributed dimensions.

2. Related Work

Structure and sequence search. BLASTp, Foldseek,² and TM-align³ provide strong baselines with explicit alignments but do not expose which embedding dimensions drive PLM-based similarity.

PLM interpretability. Prior work maps embeddings to functional labels or uses attention for contact prediction; fewer studies validate dimension-level contributions against independent per-residue structural annotations at retrieval scale.

3. Methods

3.1. *Corpus and splits*

We built a SCOP-labeled corpus of 22,118 protein chains from PDB mmCIF files. A locked holdout set of 1,500 chains (stratified by SCOP fold) was reserved before any tuning; attribution thresholds and dim→profile maps were fit on `split=train` only.

3.2. *ESM2 embedding and XQdrant indexing*

Chains were embedded with ESM2-t33-650M (mean-pooled last hidden state, $d = 1280$). Vectors were indexed in Qdrant (HNSW approximate nearest-neighbor search) with payload metadata. Queries retrieve top- k neighbors; Qdrant returns `dims_explained` (top-32 dimensions with non-negative contributions) for each hit.

3.3. *Dim→profile attribution score*

For each query–neighbor pair, we form an L1-normalized contribution vector $\mathbf{a}$ over 1,280 dimensions. A train-only map $\mathbf{M}$ correlates each embedding dimension with binned DSSP profile features (32 bins $\times$ 5 properties). The attribution-weighted prediction is $\mathbf{p} = \mathbf{aM}$. Pair scores are Spearman correlations between $\mathbf{p}$ and profile agreement vectors. Statistical tests include global row-permutation (5,000 permutations), label-permutation of fold labels, and Mann–Whitney U tests.

3.4. *Baselines and experiments*

Experiments A–F follow Table 1. Layer/pooling ablations re-embed 2,283 chains (150 hold-out queries, top-10 neighbors, train map chains) under alternate ESM2 settings and re-score attribution fold-gap on 1,043 pairs with binned DSSP profiles.

3.5. *Code and data availability*

Experiment scripts, configs, and reproduction instructions are public at <https://github.com/Moetez-Fradi/XAI/tree/bioinformatics-track/Bioinformatics>. The XQdrant fork

Table 1. Experiment overview.

Exp	Question
A	Does attribution preserve retrieval vs ESM2/BLAST/Foldseek?
B	Do attributions align with DSSP profiles (permutation tests)?
C	Do attribution scores discriminate fold match (ROC)?
D	Do attributions covary with thermostability proxies?
E	Do attributions localize to UniProt active/binding sites?
F	Workflow time and steps vs BLAST/Foldseek+TM-align+PyMOL

implementing `dims_explained` is pinned at commit `e91022ab` (<https://github.com/Moetez-Fradi/XQdrant/tree/e91022ab36c724d2110b32f004ab688fee062af0>).

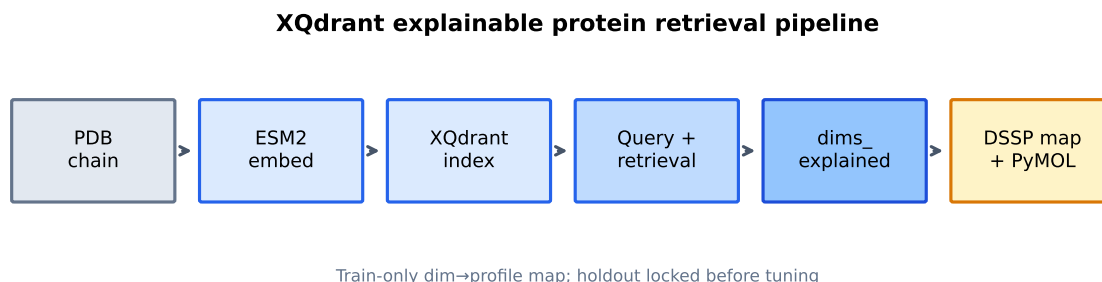


Fig. 1. End-to-end pipeline: PDB → ESM2 → XQdrant retrieval with `dims_explained` → DSSP profile mapping and structural visualization.

4. Results

4.1. Experiment A: Retrieval accuracy

On 1,500 holdout queries, XQdrant fold-level recall@1 was 0.898 and mAP 0.896 (Table 2; Fig. 2). The ESM2 cosine baseline uses exact all-pairs cosine over the same embedding corpus (R@1 0.957); XQdrant’s HNSW index incurs a modest recall gap versus this exhaustive search (0.898 R@1), consistent with standard approximate-NN trade-offs at corpus scale. Critically, enabling `dims_explained` does *not* alter Qdrant neighbor rankings (0 rank mismatches vs. plain Qdrant search on all 1,500 queries), confirming attribution is computed post-hoc rather than as a re-ranking step. BLASTp reached 0.965 R@1; Foldseek and TM-align reranking reached 0.913–0.914 R@1.

Table 2. Fold-level retrieval on 1,500 hold-out queries (Experiment A).

Method	R@1	mAP
XQdrant (ESM2)	0.898	0.896
ESM2 cosine	0.957	—
BLASTp	0.965	—
Foldseek	0.913	—
TM-align (Foldseek rerank)	0.914	—

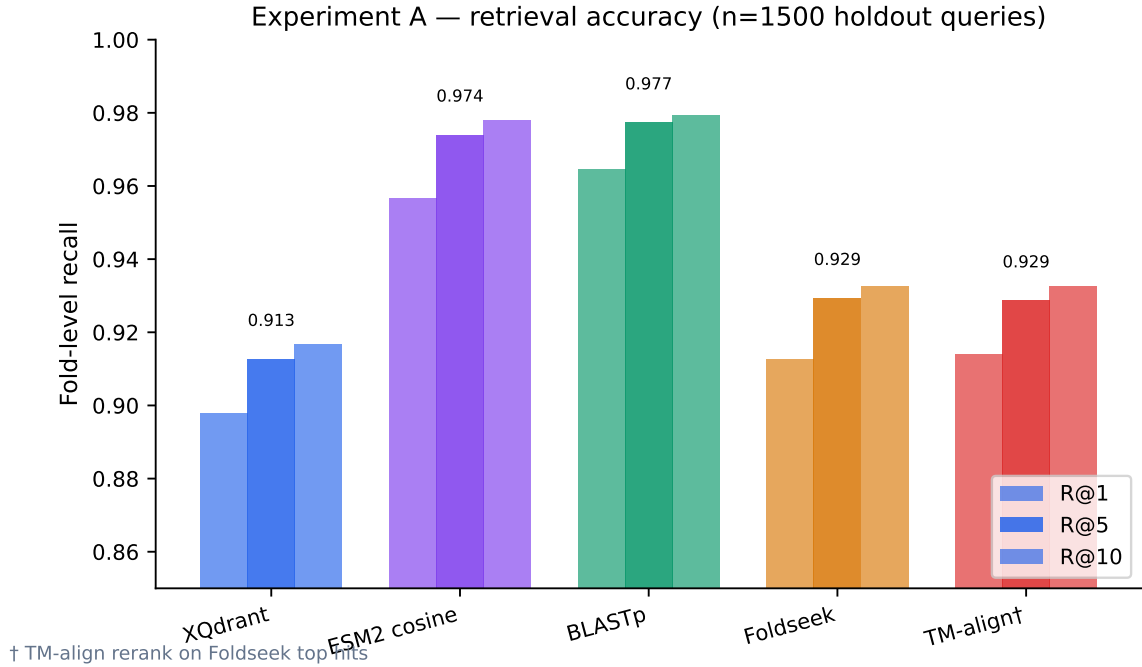


Fig. 2. Fold-level recall@1/5/10 across retrieval methods (Experiment A).

4.2. Experiment B: Attribution validity

We evaluated 10,235 query–neighbor pairs (top-10 neighbors). Fold-matched pairs scored higher than fold-mismatched pairs (mean Spearman 0.083 vs 0.037; Mann–Whitney $p = 5.3 \times 10^{-58}$; label-permutation gap $p = 0.0002$; Fig. 3). Global row-permutation did not reject the null ($p = 0.20$). Top- N dimension ablations (Fig. 4) show concentrated attributions (mean top-1 mass 0.90) with stable fold-gap across $N \in \{1, 3, 5, 10, 32\}$.

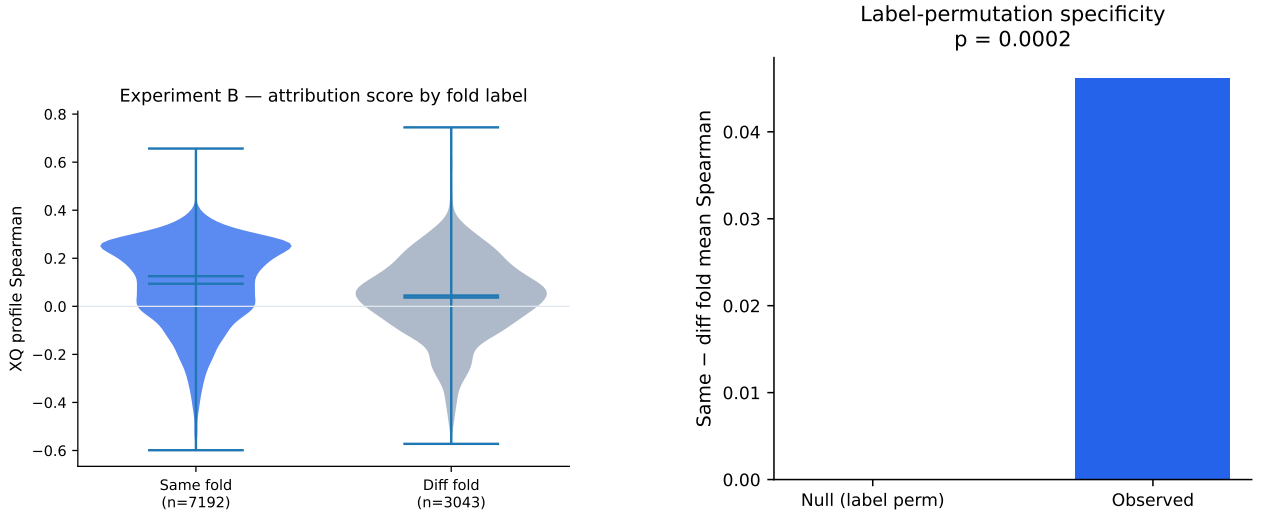


Fig. 3. Experiment B: profile Spearman by fold label (left); label-permutation gap (right).

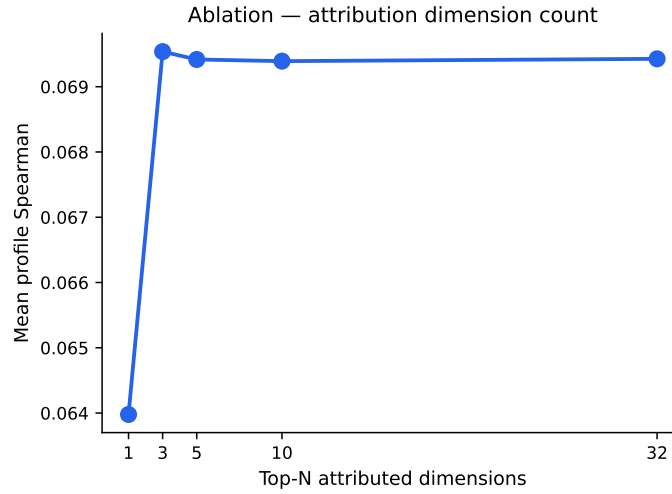


Fig. 4. Ablation: mean profile Spearman vs. number of attributed dimensions (top- N).

4.3. Experiment C: Specificity (ROC)

Attribution profile scores discriminated same-fold from different-fold pairs (AUROC = 0.63; Fig. 5), above random-dimension (0.53) nulls.

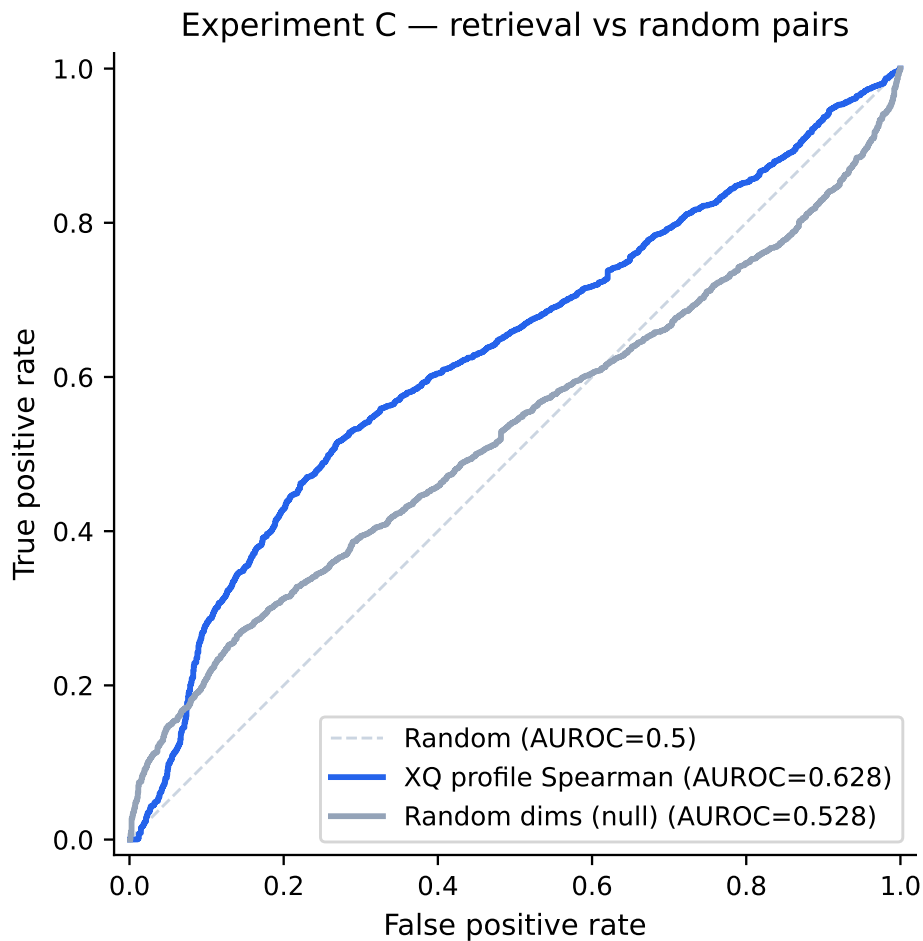


Fig. 5. ROC curves for fold discrimination (Experiment C).

4.4. *Experiment D: Thermostability case study*

On 44 mesophile/thermophile pairs, cross-pair Spearman between attribution scores and ion-pair density deltas was $\rho = -0.30$ (permutation $p = 0.052$; Fig. 6–7). PyMOL panels show RSA and charged-exposure differences on a GroEL ortholog pair (Fig. 8).

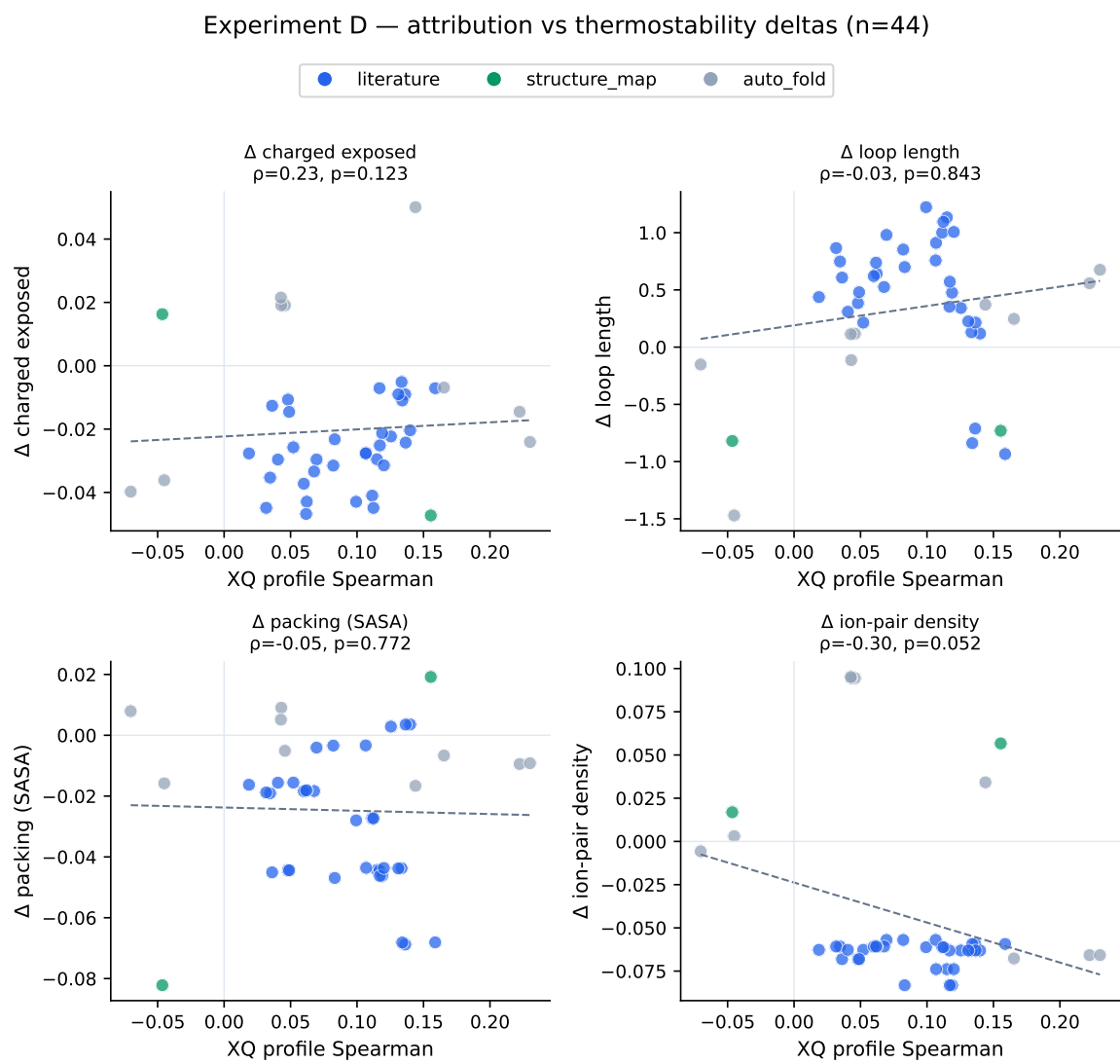


Fig. 6. Attribution score vs thermostability feature deltas (Experiment D, $n = 44$).

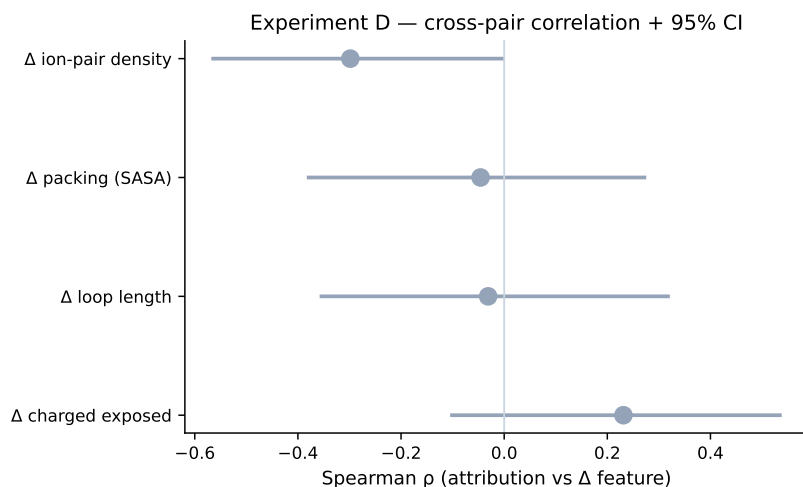


Fig. 7. Cross-pair Spearman correlations with 95% bootstrap CIs (Experiment D).

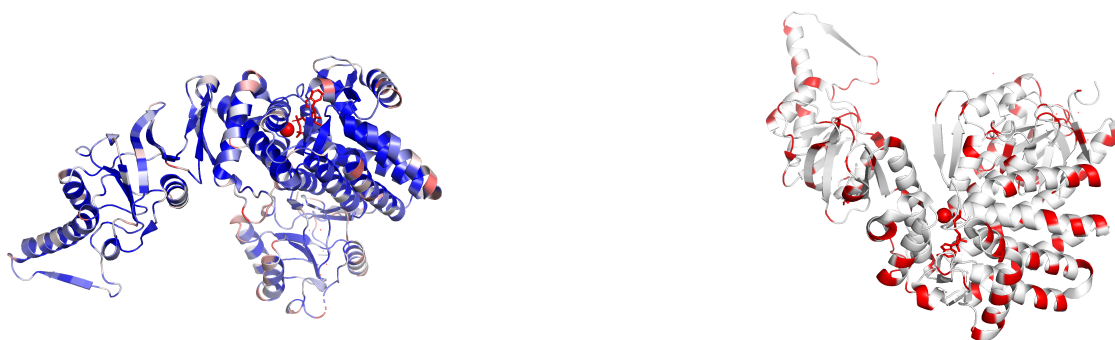


Fig. 8. GroEL meso/thermo pair: RSA (left) and charged surface (right).

4.5. *Experiment E: Active-site localization*

On 200 holdout enzymes with UniProt active/binding sites, we analyzed 1,590 query–neighbor pairs from Exp A retrieval. Per-residue XQ attributions enriched at annotated functional sites vs background (Mann–Whitney $p \approx 0$; mean site enrichment ratio 1.10 for functional-match pairs). TM-align structural overlap at sites was near unity on average (≈ 1.01) but differed significantly between functional-match and fold-only controls ($p = 2.7 \times 10^{-5}$); XQ cohort separation was borderline ($p \approx 0.05$). Fig. 9 summarizes cohort-level enrichment; Fig. 10 shows the top-enrichment functional-match query (1UWL_B) in sequence and structure.

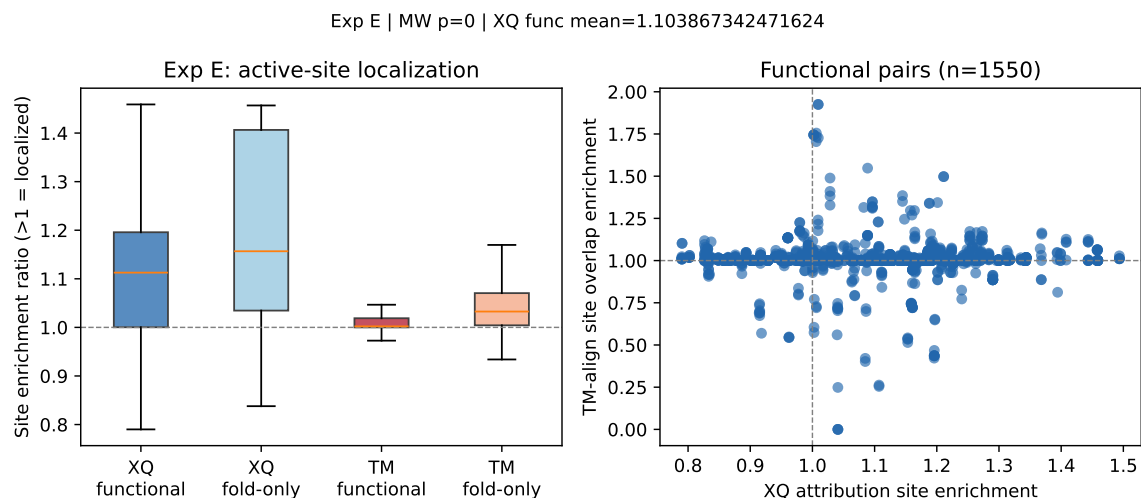


Fig. 9. Cohort site-enrichment ratios for XQ attributions vs TM-align overlap (functional-match vs fold-only controls; Experiment E).

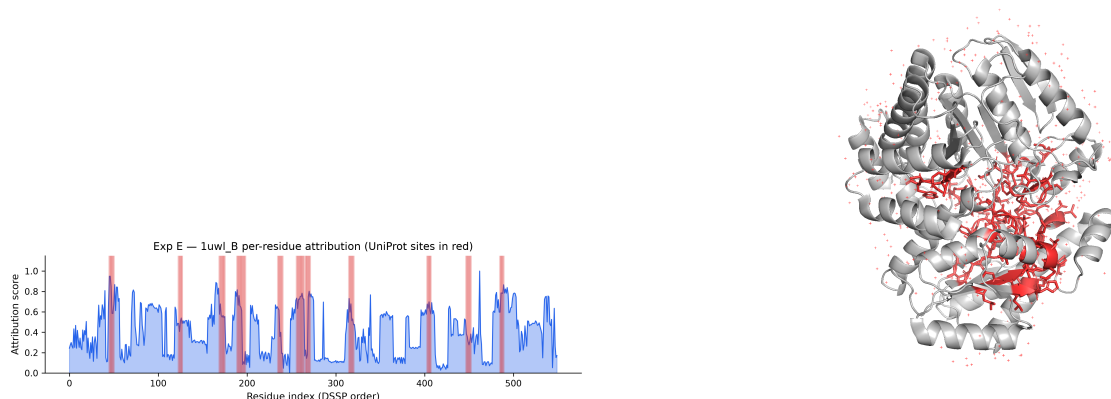


Fig. 10. Top-enrichment functional-match query (1UWL_B): per-residue attribution profile with UniProt sites marked (left) and PyMOL structure coloring with active-site sticks (right; Experiment E).

4.6. Experiment F: Workflow comparison

On 50 holdout queries, median wall-clock times were: XQdrant cold path (embed + search) 2.55 s; indexed search-only 0.074 s; Foldseek+TM-align 4.17 s automated (+10 min estimated PyMOL); BLAST+TM-align 0.61 s automated (+PyMOL). XQdrant requires 0 manual steps and returns `dims_explained` automatically (Fig. 11). BLAST+TM is fastest on automated steps alone but still requires manual structural inspection for rationale.

Table 3. Layer and pooling ablation ($n = 1,043$ pairs).

Config	Layer	Pooling	Fold gap
layer33_mean	33	mean	+0.035
layer16_mean	16	mean	−0.049
layer8_mean	8	mean	−0.112
layer33_cls	33	CLS	−0.022

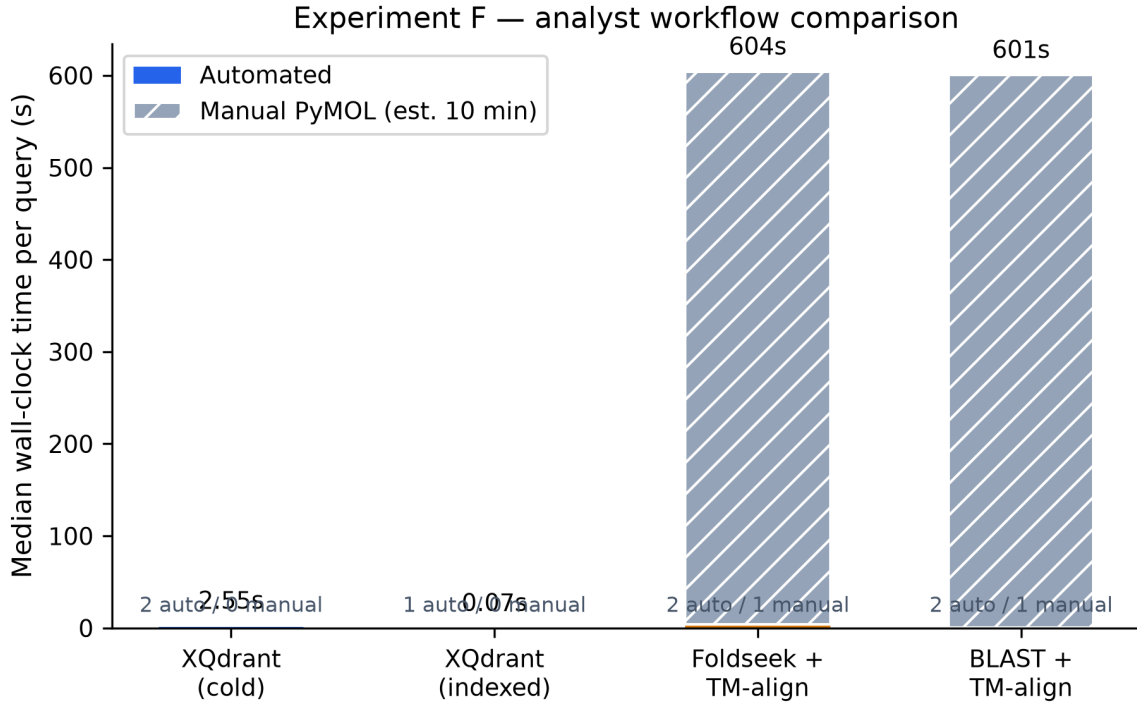


Fig. 11. Analyst workflow comparison: automated vs estimated manual PyMOL time (Experiment F, $n = 50$).

4.7. Ablations: *ESM2* layer and pooling

We re-embedded 2,283 chains under four settings and measured fold-gap (same-fold minus different-fold profile Spearman) on 1,043 Exp A pairs (Table 3; Fig. 12). Only **last-layer mean pooling** (production default) yielded a positive gap with significant Mann–Whitney separation ($\Delta = 0.035$, $p = 2.8 \times 10^{-4}$). Earlier layers (8, 16) and CLS pooling showed inverted or null specificity, supporting our embedding choice. Top- N dimension truncation (Exp B; Fig. 4) showed stable scores for $N \in \{1, \dots, 32\}$ with highly concentrated attribution mass (mean top-1 fraction 0.90).

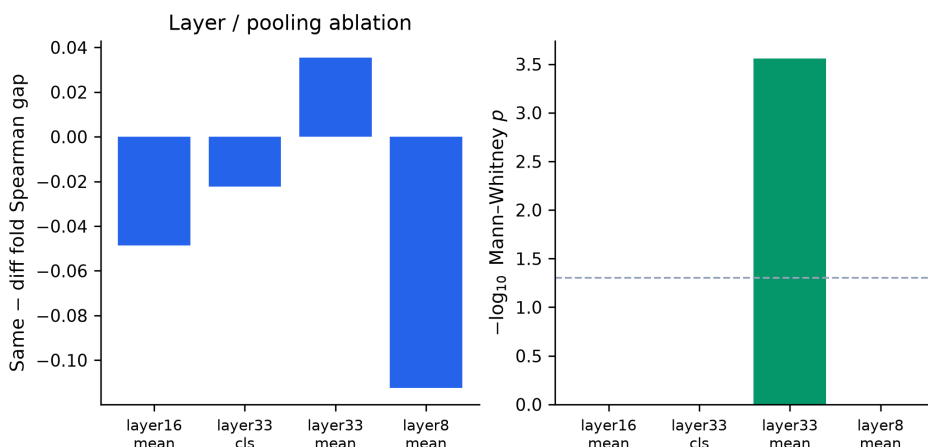


Fig. 12. Layer/pooling ablation: same-fold minus diff-fold Spearman gap. Only last-layer mean pooling is positive.

5. Discussion

XQdrant attributions are most reliable as *specificity* signals—separating fold-matched from mismatched neighbors and enriching at annotated functional sites—rather than as precise absolute predictors of global DSSP profile agreement (global row-permutation $p = 0.20$ in Exp B). Layer/pooling ablations confirm that this specificity depends on late-layer mean-pooled embeddings; earlier layers encode signal poorly aligned with our DSSP profile map. Workflow experiments show indexed attributed search at median 0.074s with zero manual steps, versus traditional pipelines that require estimated 10-minute PyMOL inspection per query regardless of BLAST or Foldseek automated latency.

Limitations. Single ESM2 checkpoint; correlational dim→profile map (not perturbation-validated); thermostability $n = 44$ with borderline ion-pair $p = 0.052$; Exp E XQ functional vs fold-only enrichment borderline ($p \approx 0.05$); ablation subset smaller than full Exp B pair corpus.

6. Conclusion

XQdrant decomposes PLM retrieval into interpretable embedding dimensions validated against DSSP profiles, functional site annotations, thermostability proxies, and analyst workflow constraints—supporting mechanism-aware screening at embedding-search speed.

Acknowledgments

References

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